Predictive Analytics of Climate Stress Impact on the Italian Mediterranean Buffalo

Physiological and Productive Responses

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Dataset Overview & Base Variables

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Dataset Overview & Base Variables

Animal Data

Total Number of Data : 2

593

300 in All Variables

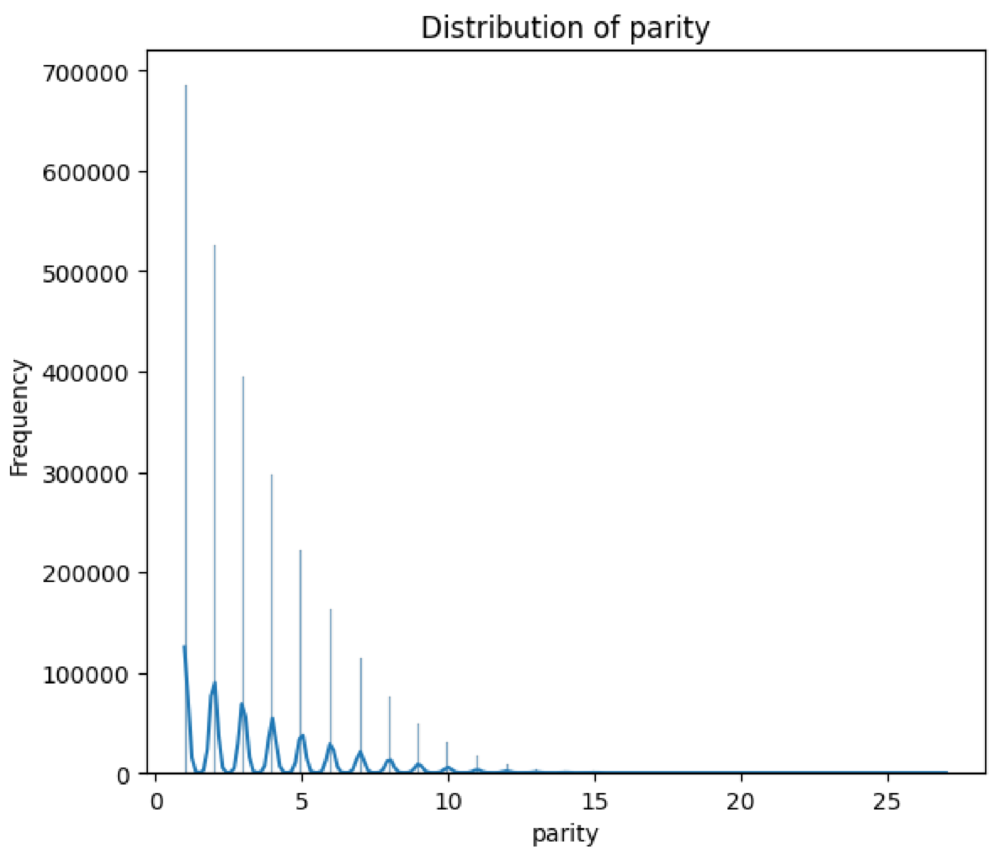
	parity	milk_kg	fat_p	protein_p	cells	NM
mean	3	90	779	457	281	2
std	2	38	239	94	798	0
min	1	0	-387	-237	0	1
max	27	886	7943	8557	98147	3

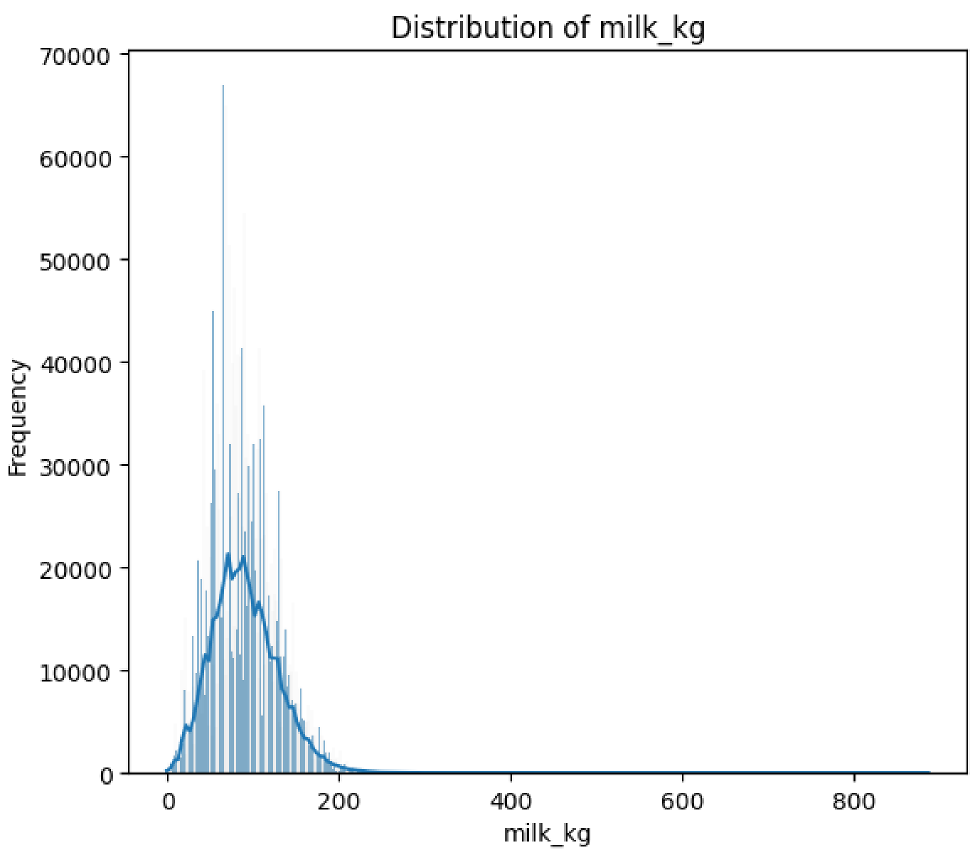
Descriptive statistics for numerical columns

Variables	Meaning
Farm_code	Identifier of FARM
Animal_ID	Identifier Animal
DTB	Date of birth of the animal
DTC	Date of calving which provided the lactation
DTT	Date of the test-date/milk quality control
Parity	Parity order
Milk_kg	Quantitiy of milk produced per day in kg during the test
Fat_p	Percentage of Fat in the milk during the test-day
Protein_p	Percentage of protein in the milk during the test-day
cells	N. of cells per ml during the test-day
NM	N. of milking event per day

Dataset Overview & Base Variables

data visualizations for numerical columns. [Unprocessed]





Dataset Overview & Base Variables

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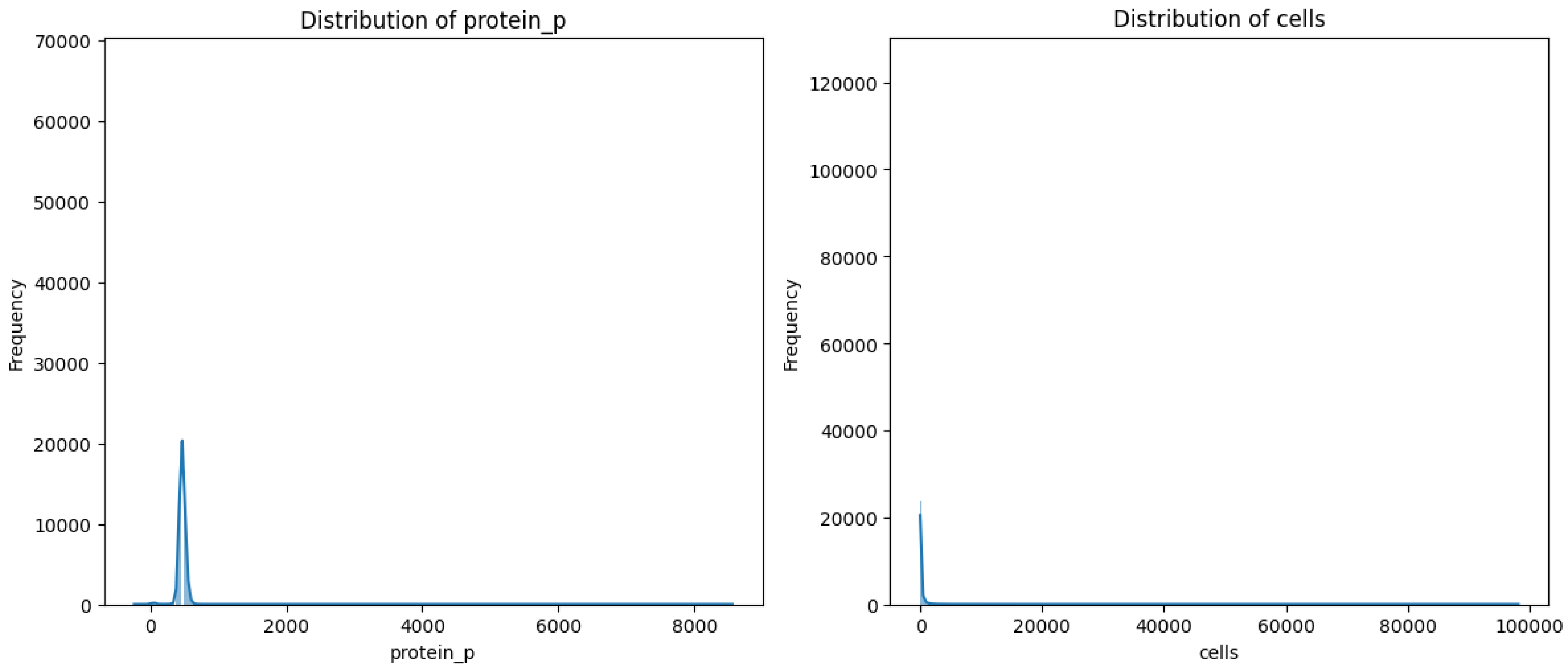

Dataset Overview & Base Variables

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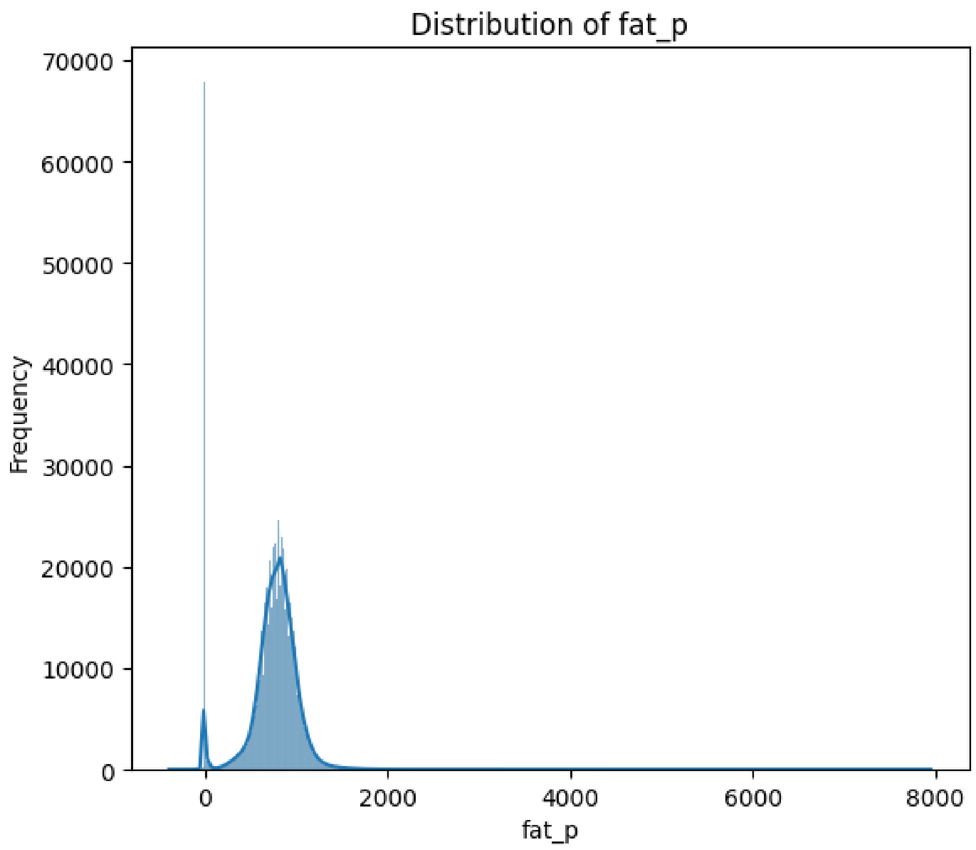

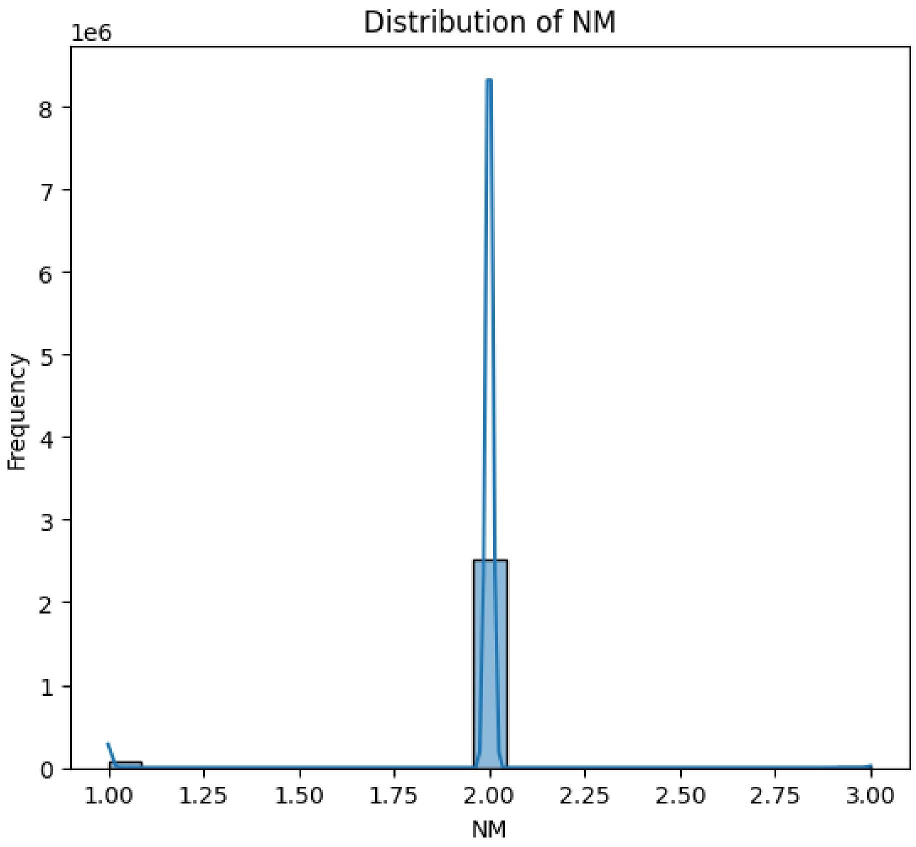

Dataset Overview & Base Variables

Farm Data

Total Number of Data : 329

Null values : 1 in Provinc

Variables	Meaning
Farm name	Primary name of the farm
Commercial name	Registered business or commercial name
Farm_code	Unique numerical identifier for the farm
ASL_code	Local Health Authority (Azienda Sanitaria Locale) code
Adress	Physical street address of the farm
City	City where the farm is located
Postal_code	ZIP or Postal code
Provinc	Province abbreviation (e.g., CE for Caserta)
Region	Italian region (e.g., Campania, Toscana)
Latitude	Geographic latitude coordinate
Longitude	Geographic longitude coordinate

Calculated Variables (FeatureEngineering)

Feature Engineering

- **Scaled variables**

$$\text{milk}_{final} = \frac{\text{milk}}{10} \qquad \text{fat}_{final} = \frac{\text{fat}}{100} \qquad \text{protein}_{final} = \frac{\text{protein}}{100}$$

- **New Variables**

Date of Calving

Date of Birth

Age at First Calving (AFC) $AFC = DTC - DTB$

Date of Test

Days in Milk (DIM) $DIM = DTT - DTC$

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Feature Engineering

- **New Variables**

  Energy-Corrected Milk (ECM) Buffalo-specific formula:

$$ECM = ([((Fat \times 10) - 40) + ((Protein \times 10) - 31)] \times 0.01155 + 1) \times milk$$

  Somatic Cell Score (SCS) Conversion to linear score:

$$SCS = \log_2\left(\frac{Cells}{100}\right) + 3$$

- **Component Thresholds**

$$\text{milk} = \begin{cases} \text{remove} & \text{if milk} \leq 0.40 \text{ or milk} \geq 26.50 \\ \text{milk} & \text{otherwise} \end{cases}$$

$$\text{fat} = \begin{cases} \text{remove} & \text{if fat} \leq 1.60 \text{ or fat} \geq 15.05 \\ \text{fat} & \text{otherwise} \end{cases}$$

$$\text{protein} = \begin{cases} \text{remove} & \text{if protein} \leq 2.67 \text{ or protein} \geq 6.68 \\ \text{protein} & \text{otherwise} \end{cases}$$

$$\text{SCS} = \begin{cases} \text{remove} & \text{if SCS} \leq -2.06 \text{ or SCS} \geq 10.73 \\ \text{SCS} & \text{otherwise} \end{cases}$$

$$\text{ECM} = \begin{cases} \text{remove} & \text{if (milk, fat, or protein)} = 0 \\ \text{remove} & \text{if ECM} \geq 30 \\ \text{ECM} & \text{otherwise} \end{cases}$$

$$720 \leq \text{AFC} \leq 1440$$

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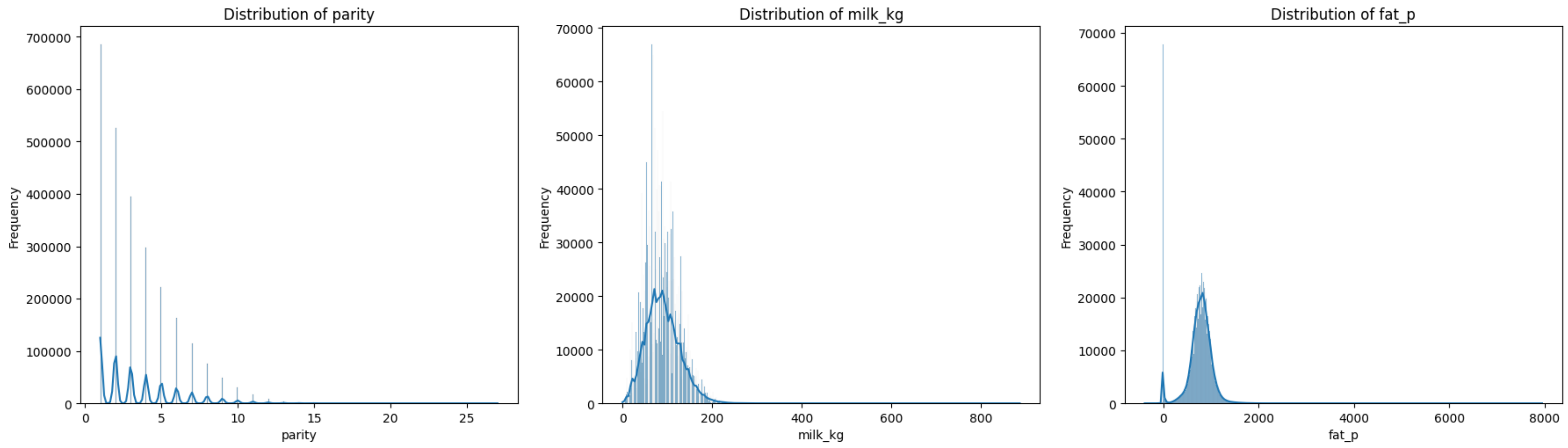
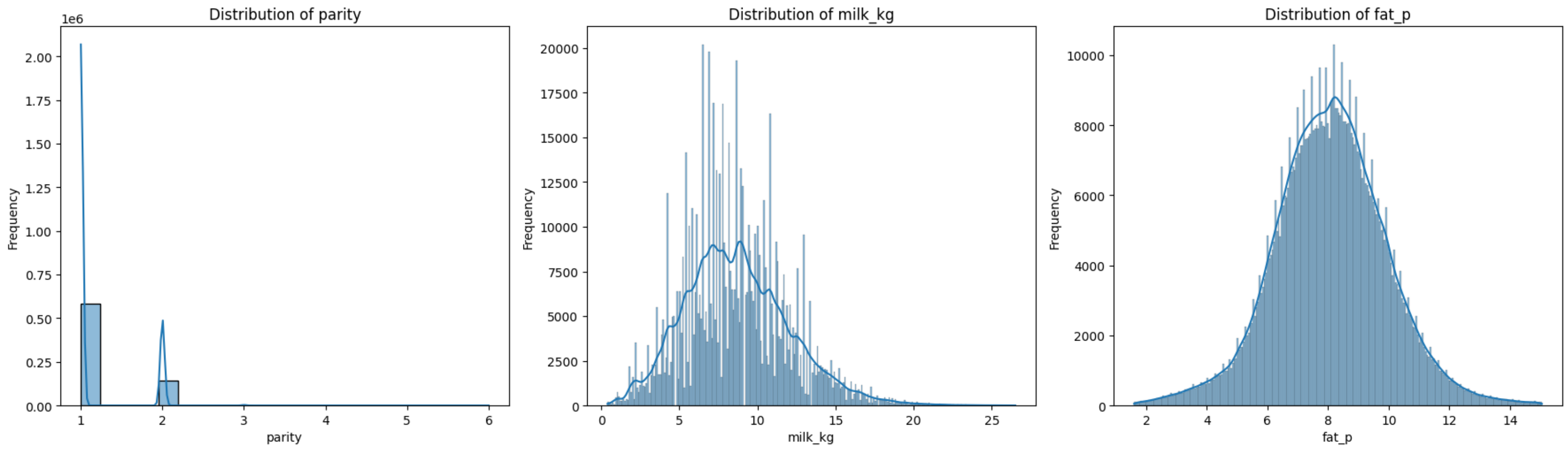
Base Data

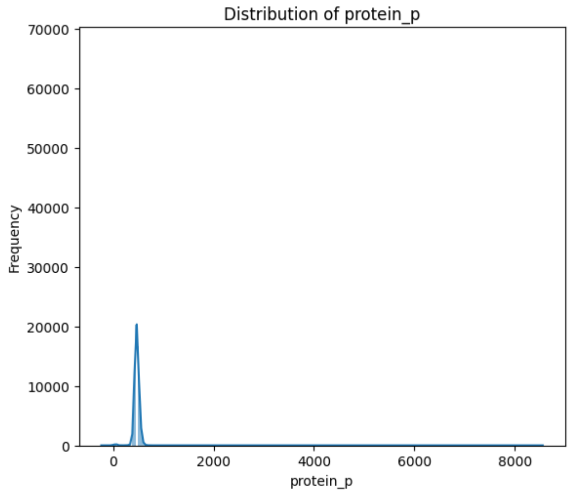


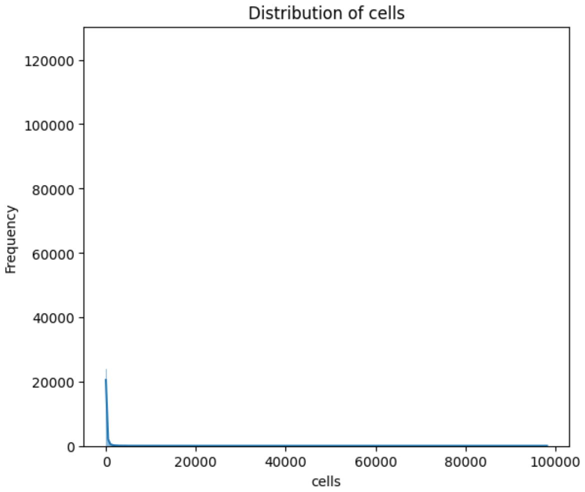
After Removing outliers

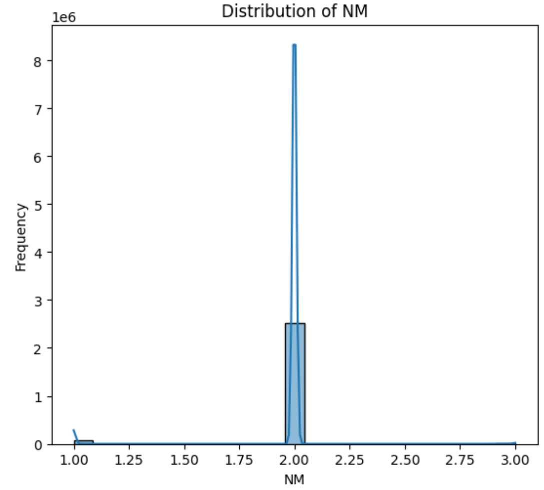


Dataset Overview & Base Variables

Base Data







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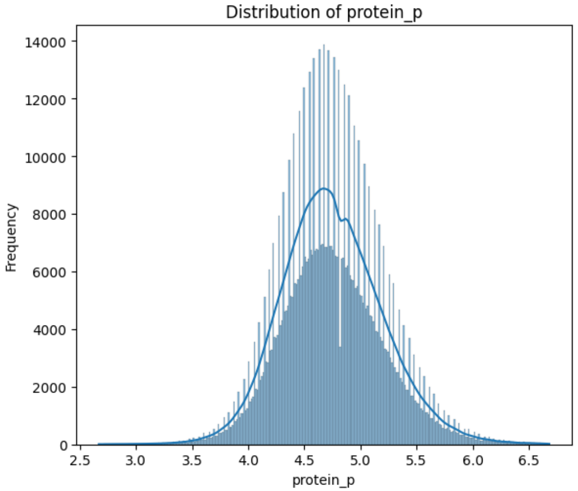

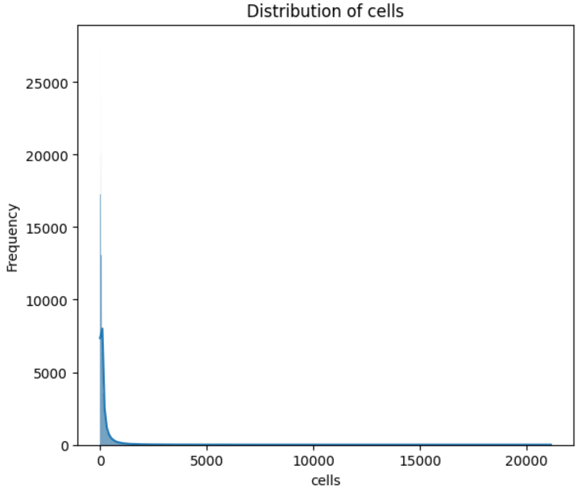

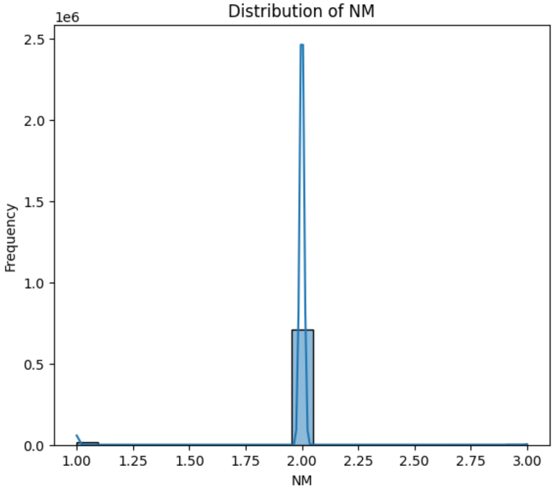

Weather Data

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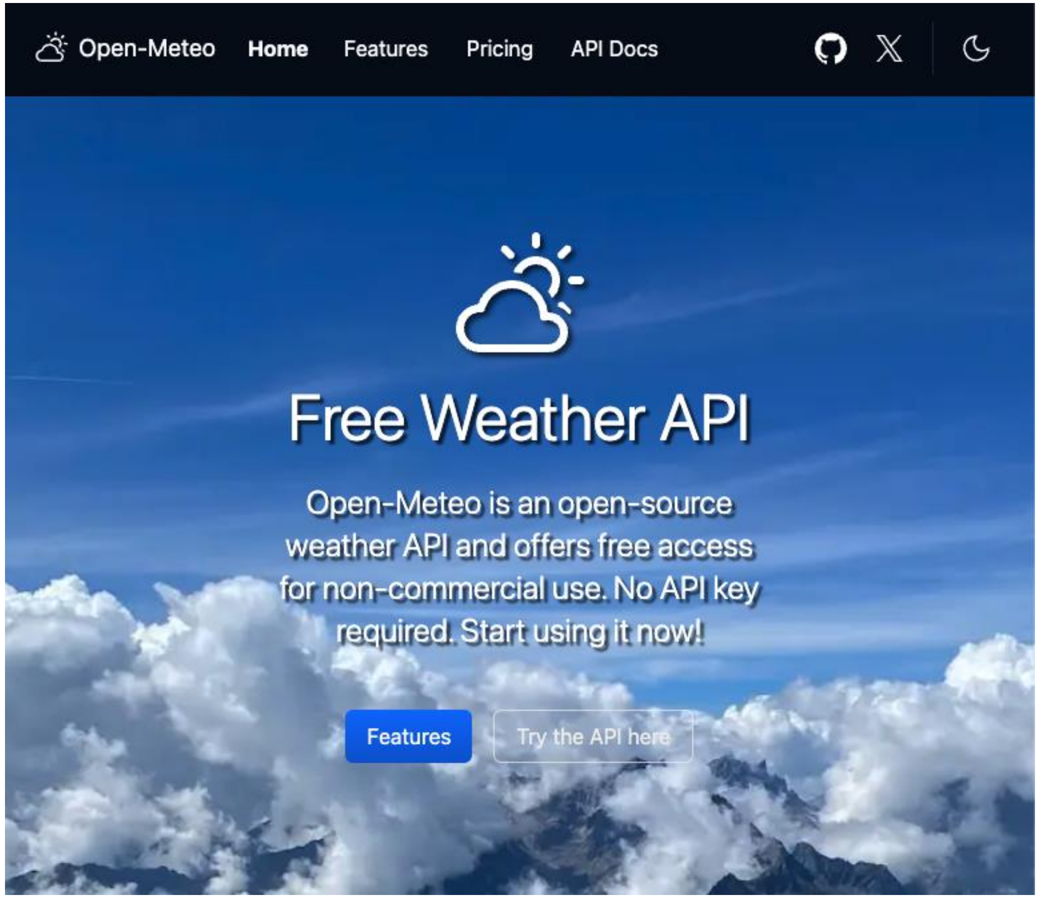
Weather Data



Open-Meteo weather API

Period : Hourly

Start Date : 2013-01-01

End Date : 2025-12-31

Location : 329 individual locations

Variables :

- Temperature_2m
- relative_humidity_2m
- wind_speed_10m

Merged Data

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Merged Data

2,593,267 rows

33 columns:

- Farm_Code
- Animal_ID
- dtb
- dtc
- dtt
- parity
- milk_kg
- fat_p
- protein_p
- cells
- NM
- ECM
- SCS
- AFC
- DIM
- temperature_avg
- temperature_max
- temperature_min
- humidity_avg
- humidity_max
- humidity_min
- wind_speed_avg
- wind_speed_max
- wind_speed_min
- THI_1_avg
- THI_1_min
- THI_1_max
- THI_2_avg
- THI_2_min
- THI_2_max
- WHI_avg
- WHI_min
- WHI_max

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